ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

The exam and the final project for A.Y. 2024-25

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The final exam

Every student has to BOTH take a written test AND present a design project.

1. The written test is individual, MUST be taken on the official test dates, and can be repeated at will until a satisfactory score is obtained. The written exam weighs 50% of the final score.
2. The project presentation is for small groups (2-3 people) (or individuals [*]), has no official test date (ask for an appointment), and can be repeated at will only by the group as a whole. The project presentation weighs 50% of the final score.

Written exams will be in presence in our labs at fixed dates.

Project presentations can be anytime and anywhere including online.

[*] Only students with ***strong*** justifications

Language of the exam

The written tests will be made available in both Italian and English, and students can choose whichever of these two languages they prefer, and even switch halfway through.

Project presentation can be had in any of the two languages. I will start in Italian with Italian students and in English with foreign students, and will switch to the other language upon request.

The written test

8/10 open text questions about the theoretical parts of the program.

- E.g.: "What is Maslow's hierarchy of needs and how does it affect the design of a product?"

Usually no exercises.

Each question has a different weight that depends on perceived difficulty and length of providing an answer.

Written tests are provided ONLY on official test dates, no exceptions.

December, January, February, twice in June, once in July, September.

Register individually to written tests, anytime you want, all the times you want.

The written exam and optional questions

Each test will have two questions clearly marked as follows:

- *Question # J (Optional for those who submitted optional exercise 1)*
- *Question # N (Required for AI students, optional for the others)*

If you wish to provide an answer anyway, please state both things: "I submitted optional exercise #n, but this is my answer: XXXX"

In this case, the question will be evaluated twice, and the grade will increase correspondingly. For instance, if the optional question is worth 3 points, then providing a 66% correct answer will correspond to 3 points for the optional exercise plus 66% of 3 points for the answer = 5 points.

Codes for this course

Are you required to follow the additional seminars on UUX and AI, and to answer the additional question? Check your codes!

90720 - USABILITY & USER EXPERIENCE DESIGN - 6 cfu

- Laurea Magistrale in Informatica – 36 hours

85573 - USABILITY AND USER EXPERIENCE (1) (LM) - 6 cfu

- Laurea Magistrale in Digital humanities and digital knowledge – 36 hours

91264 - USER EXPERIENCE DESIGN - 6 cfu

- Laurea Magistrale in Artificial intelligence – 44 hours

YES

91285 - PROJECT WORK IN USER EXPERIENCE DESIGN - 3 cfu

- Laurea Magistrale in Artificial intelligence – individual project
- You can choose these credits ***only*** if you also chose 91264, not as an alternative

Calendar of dates

During the course

Phase A deadline

- Virtuale, April 11th 2025

Phase B deadline

- Virtuale, May 6th, 2025

Phase C+D deadline

- Virtuale, 7th June 2025

Written exam

- in person, 9th June 2025 (11:00, lab. Ercolani)

Presentation of project

- in person, 10th June 2025 (afternoon, in my office, Via Ranzani 14)

After the course

- ***Written exam***

  on fixed dates, in presence:
  - One date TBD June 2025
  - One date TBD July 2025
  - One date TBD September 2025
  - One date TBD January 2026
  - One date TBD February 2026

- ***Presentation of project:***
  - either in person or online
  - by appointment starting from 12th June, 2025.
  - Last available slot: 31st January 2026
  - Reserve a slot by mail before 31st December, 2025.

Bad and good answers:
“how much is 2+3?”

Expected answer	Rejected answer
5	*Addition (usually signified by the plus symbol “+”) is one of the four basic operations of arithmetic, the other three being subtraction, multiplication and division. The addition of two whole numbers results in the total amount or sum of those values combined.* *Besides counting items, addition can also be defined and executed without referring to concrete objects, using abstractions called numbers instead, such as integers, real numbers and complex numbers. Addition belongs to arithmetic, a branch of mathematics.* *The answers therefore is a number that is higher than both 2 and 3, and it lies somewhere between 0 and 1000 .*

Bad and good answers:
“What is Maslow's hierarchy of needs and how does it affect the design of a product??”

Expected answer	Rejected answer
Maslow was an American psychologist that studied the needs of sane humans and placed them on a layered structure (a pyramid). *He asserted that humans need to satisfy lower layers first before consider higher ones. Thus for instance physical well-being comes before self-esteem or creativity.* *All design teams need to explicitly consider at which level of the pyramid the artefact needs to be placed, and organize accordingly tone, message, aesthetics and all perceivable aspects of the design.*	*A psychological study of healthy minds rather than sick ones. At the basis of everything there is physical survival, then economic and social security, the feeling of belonging to a group, the esteem of the group and others, and finally one's own actualization as a person* *Every interesting and positive object we choose addresses at least one of the Maslow needs. Therefore, does our product respond to the fundamental needs of survival and security? Protect the user from threats or other issues? Provide to the needs of an individual, or of a larger social unit, such as a family or a group? Promote a sense of belonging? Promote self-esteem or a sense of personal accomplishment?*

Copy-pasted from the slides

The project

You need to implement a group project (2-3 people). Individuals only in case of emergency or provable logistical difficulties.

Today I am presenting the requirements for the end-of-course project.

There is also a specific implementation model that you have to follow and that will be presented on the same day.

No programming required, nor expected. This is a design course, not a technological one. Nonetheless, if you show up with a working prototype, I will not complain.

Two modes for the project

During the course

The project is split in three phases

- Phase I (specifications today, deadline late March): ***Ethnographic assessment***
- Phase II (specifications in early April, deadline early May) ***Personas***
- Phase III (specifications in early May, deadline at the exam): ***design and evaluation***

Pros and cons

- Pros: you finish the exam early
- Cons: you have to start working immediately and keep on working.

ONE exam date in early June

- Project presentations and written exams.
- This is the only mode for students leaving Bologna in June
- All other students are welcome to choose it, of course

After the course

The project is a single activity

- It has the same components as the other one, ***Ethnographic assessment, Personas and Design and evaluation***
- Can be prepared any time between now and end of January 2026.

Pros and cons

- Pros: you can breathe and work at your pace.
- Cons: you have to wait for exam sessions (more or less).

No access to end of May date

- You have to wait for other regular dates
- Applies to both project presentations and written exams.

The project presentation

You need to submit a set of documents on the day of the presentation. No advance submissions. No software.

- The number and names of these documents is predefined and strict

We then meet and discuss their content.

- ***First part: the process –*** what steps did you perform to obtain the result you obtained
- ***Second part: the project –*** can you convince the client to finance the design/redesign of their tool?

All team members are expected to contribute equally to the activities and work on all parts.

Divide tasks horizontally, not vertically.

The 2025 UUXD project: FRTR

From Reluctance To Routine

FRTR is the design/redesign of a web site / web application / mobile app / desktop application aimed a very specific audience that ***must*** use your tool but is suspicious or afraid, and you need to buid trust

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From reluctance to routine

Understanding your audience and providing for a specific need or goal of your audience is a key for a successful product.

- *Products* (including applications, mobile apps and web sites) are succesful if they are designed for a clearly understood and specific audience.
- *Forced products* (including applications, mobile apps and web sites) are those that are the only available choice to carry out some task in a digital form. They are succesful if they they are able to be perceived as worthy and preferable to carrying out the same task by hand. This means making them clearly understood by a specific audience.

The impact (given by words, images, layout) is the main means of the delivering the reasons and ways to use proficuely your product.

From reluctance to routine

- The client is a large organization that provides an online portal for citizens/users/customers.
  - e.g.: senior citizens to manage healthcare, pensions, and other services.
- Despite the platform's usefulness, many users avoid it due to digital anxiety and unfamiliarity. This leads to increased costs for in-person assistance and delays in service access.
- The client seeks a solution that transforms the tool into an intuitive and empowering experience, helping users develop long-term digital confidence. You are tasked to improve the user experience of such an existing service.
- Each project will identify a specific situation, in a specific domain, and ***a very specific audience***, and create a web site / web application / mobile app / desktop application that supports the specific needs and goals and tasks of this audience.

From reluctance to routine

- The tool must integrate clear information with accessible services, designed to guide users from initial hesitation to regular, confident use. The redesign should reduce friction, instil trust, and foster a sense of progress, encouraging frequent and sustained interaction with the tool.
- Careful and empathic understanding of their specific characteristics will be important. Today I will suggest some domains, tasks and user types. I will also propose some functions and models.
- You can choose one of them, or suggest your own. You can make proposals (under some rules).

From reluctance to routine

You can suggest a domain+task+audience for your application, as long as you follow these steps:

- It addresses a very specific (and specified) domain
- It addresses one or a few of very specific (and specified) tasks
- It addresses one very specific audience, well identified and described.
- It shows the audience is ***not free to use a different tool.***
- It shows you did your best to study, understand and empathize with this audience. You must evaluate how well you did.
- It clearly needs to provide a message to help/convince/assist that specific type of user to use it in carrying out tasks.
- The audience is not using your tool yet, and is skeptical. You must inform it of its existence, usefulness, appropriateness.
- Please try to go beyond information sites, social networks and e-commerce tools.

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Objective

The tool must include both information content and active services.

The redesign must allow the company to address and convince ***a specific audience*** to increase the use of the tool and therefore increase the profit/reduce the costs/ improve the services it provides.

Assume that:

- The organization is large and complex.
- It has money, but uses them wisely (it does NOT overspend).
- It has a mature web presence, and handles much of the standard chores of the website already (brand recognition, user management, FAQ, legal texts, etc.)
- The new redesign must improve/replace (some parts of) an existing tool and be presented as a stable and immediately recognizable re-organization of the old tool.
- The audience is neither too wide neither too narrow, and it can be well defined with just a few words.

Careful and empathic understanding of their characteristics will be a key evaluation factor.

Objective (2)

- You do not work for the above-mentioned client.
- You work for a software company that provides services and tools to this and other clients.
- Your company has to convince the client to invest in the tool and it must provide a convincing case for the budget.
- You first have to convince your project manager, then the client itself.
- Do not design a product. Do not design a web site. Do not design an app.

Design the user experience of the web site/app!

Background

The lack of confidence lies in the nature and interfaces of the tool, ***not*** in:

- An inferior product/service
- An unknown brand
- A specific unawareness of the existence of the tool

The solution must revolve around the awareness, acceptance and attractiveness of the redesigned tool, and ***not*** around:

- traditional advertisement campaigns
- Walls of text
- Mere simplification of the vocabulary used by the interface (i.e.: different labels, more explanations, etc.)

You ***can***, however:

- Target the language(s) of the site
- Target the structure of the site and of the pages

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The target audiences (examples)

1. Tool aimed at parents of kids (5-12 y.o.)
   Although the information/services/ it provides are of good quality and correct, parents still prefer to call the school or a teacher, and use it as little as possible and only helped by operators.
2. Tool aimed at foreign immigrants
   The products/services are fine, and target users provide all the services they need to carry out the needed tasks, but they are not sure, they do not understand, they prefer to use agencies, intermediareies, or a long queue in a physical office.
3. ~~Tools for smart university students 19-25~~
4. Tools aimed at retired elderlies
   Users have the need for the product and services, have rarely used online or interactive tools to get them, and do not see any particular reason to switch to digital. The current tool does not help at all.
5. Tools aimed at travellers
   Both business travellers, tourists and commuters. Not just tickets, any service/product that can be of interest. Why don't they use the tool? What could be done to improve this?

1. Tool per genitori di bambini (5-12)
2. Tool per immigrati stranieri
3. ~~Strumenti per studenti universitari~~
4. Tool per pensionati anziani
5. Tool per viaggiatori

Common aspects of all projects

You must design a site/subsite/mobile app of a more motivating version of the existing one.

The tool must provide ***both content and services***.

It must target ***the selected type of users*** very precisely. All of these services DO NOT cater for a generic Mario Rossi user: they all address very special users that YOU have to identify, define, describe.

You have to identify the reason for their lack of trust/confidence in the products/services/ tool and address it adequately and precisely.

You have a large freedom of design decisions within this framework. Use it! Be creative!

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Design fiction

- Use whenever possible narrative and dramaturgical approaches to analysis, design, evaluation, documentation.
- Narrative approaches must have a direct and explainable connection with specific aspects of the analysis, design, evaluation that took place, and you can show exactly how this connection is not trivial but actually meaningful. Please document correctly the interaction between the narrative and the actual design.

- This year's project regards specifically ***confidence***. You must:
  - Consider target users not trusting the site;
  - Consider target users not confident in their own ability;
  - Consider target users needing to verify the correct and timely completion of their tasks.

Evaluation of the project

- Functional aspects
  - Credible design of the content and services designed for the audience of the system
  - ***You can assume completed the design of common features (e.g. registration, login, forum, FAQ, etc.) therefore just focus on features specific for the needs and goals of the main audience.***

- Non-functional aspects of the main features
  - Credible design of the usability and user experience details of the system for the main audience of the system.
  - ***Be creative. Be bold. Be daring. Surprise me.***

Assessment Criteria

Your project will be evaluated taking into account seven criteria. You can obtain 5 points for each, for a grand total of 35.

1. Relevance

   Is the tool addressing the right problems?

2. Coherence

   How well does the toolfit with respect to similar external services?

   Is it internally coherent?

3. Effectiveness

   Is your redesign actually achieving its objectives?

4. Feasibility

   How well are resources being used?

Assessment Criteria

Your project will be evaluated taking into account seven criteria. You can obtain 5 points for each, for a grand total of 35.

5. Impact

   What differences does the tool make to the intended audience?

6. Sustainability

   Will the benefits last?

   Is the tool easy to maintain and expand to new features?

7. Persuasiveness

   Is your design ultimately convincing?

Relevance

Is the project addressing the right problems?

The extent to which the project objectives and design respond to stakeholders/clients needs, policies, and priorities, and continue to do so if circumstances change.

✓ **Do's**:

- Be clear about the objectives of the service you plan to design.
- Be clear about the needs and the goals of the user for whom you are designing.

🚫 **Don't's:**

- Designing generically and loosely while trying to please everyone.
- Avoid choosing one specific user.

Coherence

How well does the project fit with respect to similar external services? Is it also internally coherent?

The compatibility of your design with other competitors that answer to similar questions.

✔ **Do's**:

- Consider the context of the service being designed. Match between system and the real world (Nielsen's *Second Heuristic*).
- Be consistent and keep the same language throughout the system to avoid confusion.

🚫 **Don't's:**

- Reinvent the wheel.
- Confuse extremely sophisticated and clunky design with clever solutions. If you haven't seen it before, there's probably a good reason.
- Designing (atomic) elements of your system without regard to how they connect to others.

Effectiveness

Is your redesign actually achieving its objectives?

The extent to which the intervention achieved, or is expected to achieve, its objectives, and its results, including any differential results across groups.

✓ **Do's**:

- Test your design comprehensively and extensively in order to obtain enough data for claiming tangible improvements.
- 3 I's: Iterate, iterate, iterate.

🚫 **Don't's:**

- Test your design with (one or more of the following): very few (e.g., < 3), biased (e.g., loved ones), imaginary (i.e., not real), out of target users.

Feasibility

How well are resources being used?

The extent to which the intervention delivers, or is likely to deliver, results in an economic and timely way.

✓ **Do's**:

- As the name suggests, a feasibility study is crucial!
- Prefer simple tools (e.g., pen and paper) to sketch simple but efficient mockups rather than using complex solutions that cost time and energy (and will probably be discarded by the client nonetheless).

🚫 **Don't's:**

- Propose fascinating but hardly believable or shockingly expensive (in terms of time or resources) features for your system.
- Put aesthetics above functionalities.

Impact

What differences does the project make?

The extent to which the intervention has generated or is expected to generate significant positive or negative, intended or unintended, higher-level effects. Does the service positively change the way users relate to it? Does it promise and achieve a change in perspective from a community?

✓ **Do's**:

- Think about the effect of your redesign in the long run. E.g., does it help to improve traffic, build confidence, promote trust?

🚫 **Don't's:**

- Redesigning a service that was already very bad or pointless to begin with.

Sustainability

Will the benefits last?

The extent to which the net benefits of your redesign will continue, or are likely to continue.

The ease of extend the tool to new and more advanced services and content

✓ **Do's**:

- Design thinking about flexibility, scalability and reusability.
- Help users recognize, diagnose, and recover from errors (Nielsen's *Ninth Heuristic*).

🚫 **Don't's:**

- Avoid a long and convoluted user journey (Information Architecture is your friend!)

Persuasiveness

Is your design ultimately convincing?

- For better or for worse, design is also (and mostly) a critical and interpretive effort, an ongoing dialogue between humans, machines and their world. There is not always a best choice at hand, and one cannot always stick to the plan or follow guidelines. Therefore, design is properly neither an art nor a science. We can call it a *rigorous discourse*, or a knowledge that can combine a certain amount of creativity with analytical elements.
- For these reasons, an evaluation criterion will also be the *persuasiveness* of your project, that is, your ability to convince the client of the quality of your design. This is not an absolute criterion, nor can it be reduced to your rhetorical ability, but rather it is the set of insights, creative and out-of-the-box solutions that you used to produce a persuasive design.

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Project Management

Documentation

- The design is only half of the work. First you have to plan the design and the evaluation, and to document it appropriately
- You have to conduct the project in a controllable and repeatable way following precise guidelines in phases, activities, outcomes, etc.
- A good product with a bad documentation will be evaluated WORSE than a bad product with a good documentation.
- Documentation needs NOT be extensive (I have seen hundreds of pages in documentation. That is too much and no-one will read them)
- I am providing documentation based on fixed form templates organized in separate cards.

- ***Templates available at*** **`https://bit.ly/UUXProject2025`**

Project phases

Phase A

1. Ethnographic research
   1. Business purpose
   2. Segmentation
   3. User research
2. Assessment of existing resources
   1. Expert Usability Review
   2. User testing

Phase B

3. Feasibility Study
   1. Context of use
   2. Scenarios
   3. Personas
   4. Recommendations for design

Phase C

4. Design Proposal
   1. Information Architecture
   2. Interaction Design Approach
   3. Structure Blueprint
   4. Wireframes

Phase D

5. Evaluation of design
   1. Inspection
   2. User testing
6. Final recommendations

Note

- Project must be carried out in order: A, B, C, D.
- Evaluations will be in a different order: A, B, D, C.

Phase A

1. Ethnographic analysis

Explore the users and the services associated to the users

1.1 Business purpose

1.2 Segmentation of target audience

1.3 User research

Please notice that the users are NEVER generic. This year I will not accept sentences like:

"Users are between 18 and 65 years old"

Phase A

1.1 Business purpose

The client:

- Who is the client and why should he want to finance a new development project?

Product Objectives:

- what do we want the client to get from this product?

Business Goals:

- How does this specific tool integrate with overall business goals?

Brand identity:

- How does this specific product promote a brand identity and make it more appreciated?

Success metrics:

- When and how do we know that the project is completed and it went well?

Phase A

1.1 Business purpose (2)

The ideal client has a specific tool, a specific audience, adequate budget and no internal resources to carry out this project.

Choose your clients carefully!

Bad clients

- Client is too big (Apple Computer, Coca Cola, the United Nations, etc.).
- Client is too small (a shop in your town, a small volunteer association with no employees, a free app created by an individual, anything existing that was not created by a professional team)
- Scope is too wide (all digital services of a large company)
- Scope too narrow (a single section of a functioning tool)
- No budget available

1.2 Segmentation of the target audience

Phase A

Often the number and variety of expected user makes it difficult to create a homogeneous characterization of them

Segmentation allows you to group users into subgroups that are homogeneous for some features

Demographic Segmentation:

- Age range, education, marital status, income, residence, etc.
- Very generic (18-59 years old men) or very specific (women 25-35 of the South graduates and unemployed, unmarried, income <20,000 euros per year)

Psychological Segmentation:

- Personality, values, attitudes, interests, lifestyle, etc.
- Motorcycle single engine mechanic

1.3 User research

Phase A

Market research methods

- Survey, focus group,
- They are indirect sources of data collection

Contextual inquiry

- Interviews, field observations, passive presence
- They are direct sources of data obtained by interacting with users

Task analysis

- Identify the context of tasks where the user uses, uses, or wants to use the product.
- The aim is to observe the sequence of steps and user interaction between the space covered by the product and other tools / events / activities external to the tool

Phase A

2. Assessment of existing resources

Find one or more application ideas you may want to use as a driving model for your application.

Find the web site of a company that could find the extreme customization subsite appealing.

Perform on them the following steps:

3.1 Expert Usability Review

3.2 User testing

2.1 Expert usability review

Phase A

The expert usability review is an analysis of a system without the help of users, on the basis only of general guidelines (eg guidelines for the web), and not driven by users' description.

Before the inspection

- Collection and justified adoption of guidelines

Inspection

- First system exploration
- Direct Analysis: System vs. guidelines
- Reverse Analysis: Guidelines vs. system
- Detect and describe a reasonable number of errors

After the inspection

- Identification and description of the tasks provided by the system
- Identification and description of user target

Phase A

2.1.1 Choice of guidelines

- ***Before*** the initial inspection you must identify guidelines that are to be used for the analysis. For instance they may concern:
  - Minimization of the cognitive load of users
  - Visibility of the system
  - Control of the interaction
  - Learnability
  - Subjective satisfaction
- The guidelines must be numbered so that they can be recalled in the evaluation sheet.

Phase A

2.1.2 First Inspection of the System

First together, and later separately, all group members explore the system to understand the main features of the application, including:

- The services offered
- The target users
- Any problem that can be immediately identified

This initial inspection is useful to quickly identify some key problems of usability and the goals that the system appears to satisfy.

2.1.3 Direct analysis: system vs. guidelines

Phase A

Systematic exploration of the application with respect to the guideline.

- Evaluation based on the functions avalable within the system
- The evaluation is independent of expected audience and their tasks
- Systematic analysis, screen by screen, of which guidelines are violated, how often and with what impact.
- In case a problem you have found is a breach of more than one guideline, list them ALL, in order of severity or causality.

2.1.4 Reverse analysis: guidelines vs. system

Phase A

Systematic exploration of guidelines with respect to the system.

- Evaluation based on the structure of guidelines
- The evaluation is independent of the expected audience and their tasks
- Systematic analysis, guideline by guideline, os which functions breaches it, how often and with what impact.
- In case a problem you have found appears on more than one function / section of the site / interface, list them ALL, in order of severity or causality.
- Obviously many of the problems found in the direct analysis can be found in the reverse one. Ignore them and look for new problems.

"A reasonable number of errors"

Phase A

How many are "a reasonable number"? No simple answer, but

- All implementation, catastrophic, and severe implementation problems have been identified, 75% of minor mistakes and many cosmetic mistakes
- All of the frequent and/or persistent problems have been found, regardless of their impact.

Each error is described in the documentation:

- Description of the function / section / interface in which it is present (or all if persistent)
- Description of the actual problem
- How the problem was discovered (before exploration, direct analysis, inverse analysis, user testing (later))
- Classification of the error (eg according to Persistence, impact, and frequency)

Phase A

2.2 User testing

- Before:
  - Define a testing protocol
- During
  - Carry out all tests
  - Collect data
- After:
  - Analyze subjective and objective data
  - Identify usability problems
  - Integrate with the previous list
  - Create curves of urgency
  - Write conclusions and suggestions

2.2.1 Definition of the testing protocol

Phase A

- Testing method (e.g., discount or full usability testing)
- List of tasks,
- Number of tests,
- Number of subjects,
- Testing methodology (eg, Thinking Aloud or Bottom Line Data)
- Metrics of tests (according to EEE)
- Recruitment of real subjects and relationship with target audience
- Organization of tests, initial introduction document, and final assessment questionnaire.

2.2.3 Analysis of subjective and objective data

Phase A

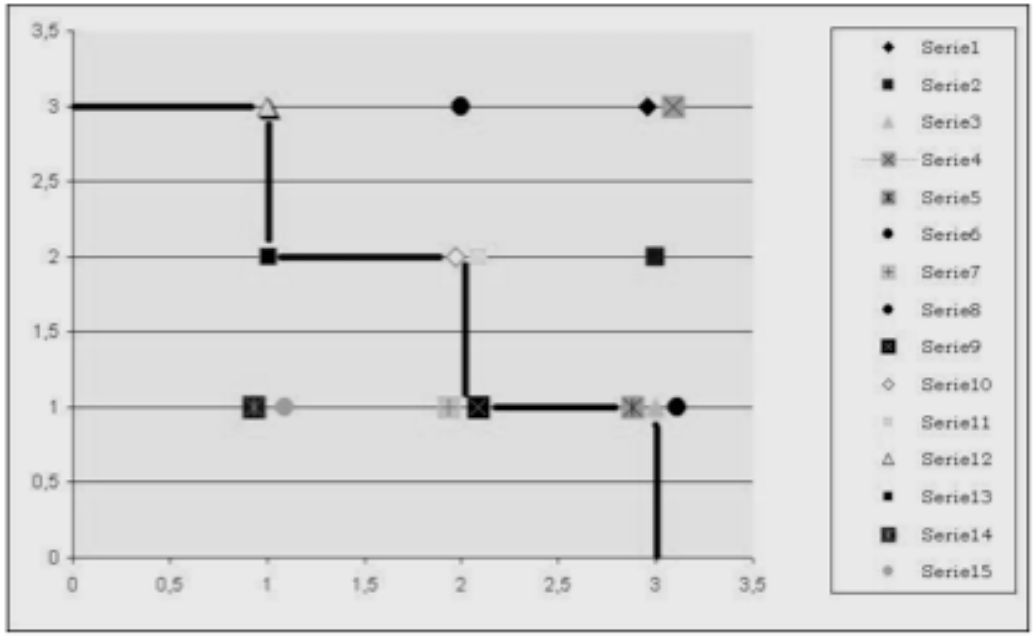
Tests will generate to a large amount of data that should be organized and described. Eg:

- "All users have all complained of a unclear layout of the information and little refined search tools.
- "For the task one needs to know the right section in advance, otherwise the information can only be found by browsing though all the sections or through free search, carefully avoiding the first result that is always "a complete set of kitchen accessories" .
- "Users first searched for a contact section or email addresses. Someone tried the "showcases" section. The only one of the three test subjects who managed to complete the task succeded by chance.
- "No one was able to complete the task"

2.2.4 Urgency Curve

Phase A

- The curve of urgency is a two-dimensional chart such as "impact vs. frequency" or "impact vs. persistence".



- All identified errors are drawn on the chart.
- Deciding on a urgency threshold (*above*: fix immediately; *below*: next release) and draw the curve separating the urgent problems from the others.

Phase A

Two issues

- A common misconception is to associate a urgency level as soon as the error is identified.
  - The team can identify only the persistence of the problem, but frequency and impact are determined only by usability testing.

- A second common misconception is to organize the usability test as a way to test the errors detected by the team
  - The usability tests verify the frequency and severity of errors with regard to a user's task.
  - For instance, a gross violation of the guidelines in a sector of the site where there are no major task is not a serious error

Info for Phase A submissions

Phase A

- The deadline for Phase A submissions will be

  Sunday 11th April 2025, 23:59

- All submissions will be made using virtuale.unibo.it in a proper area that will be created in the next days.
- Only submissions delivered by the deadline and followed by two more submissions (phase B, and phase C+D together) on the corresponding deadlines will make the authors eligible to participate to the written exam of early June. All other dates are open access.
- A number of documentation templates are made available on Google Drive. June students ***must*** use them, all other students may or not according to their tastes.
- ***Templates available at*** **`https://bit.ly/UUXProject2025`**

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Info for Phase A

Phase A

- ***Team card***
  - Your identities
- ***Client card***
  - fill in all fields. Mainly text. Do not be too concise, do not be too verbose. There are indications of expected size.
- Ethnographic analysis

  User segmentation card, market research cards
  - Decide a small number of segments you feel are more important than the others. Describe them and justify your choice. Be specific and convincing
  - Use one or many of the suggested methods to do user research: available statistics, online surveys, focus groups, in person interviews, field observations, passive presence.
  - List your non-trivial design ideas after the conclusion of this phase

Info for Phase A

Phase A

Assessment of existing resource

- Evaluate the current tool + one or more competing or inspiring alternatives (may be different purpose but show ideas you would like in your tool)
- ***Assessment card***: fill in one card for every tool you evaluate
  - ***First inspection***: use and play with the tools for a while, trying to understand (your ideas of) strong points and weak points. Prepare a lot of screenshots. Document your conclusions in a single page text, maybe as a list.
  - ***Direct and reverse analysis***: use the User Focus guidelines and the corresponding tool (an Excel file).

    http://www.userfocus.co.uk/resources/guidelines.html

    Perform direct and reverse analysis on all tools. Include the quantitative results in final documentation, include the full file in the accompanying folder.

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Info for Phase A

Phase A

- User testing ***(Preliminary user testing card)***
  - Recruit 2-6 people (***real*** people), possibly outside of the inner circle of your peers (may be difficult).
  - Ideate 3-4 tasks (full tasks, not single actions, meaningful for the user). Try to precise and concrete
    - good task: *find the cheapest laser printer in catalogue*;
    - bad task: *examine the available laser printers*.
  - Testing protocol section: details about the test.
  - Test data card for every subject to test
  - At the end of the tasks always carry out a System Usability Scale test (template available on shared folder) and an appreciation questionnaire

  - ***Important***: inform at least ***TWO*** of the above people that you shall need their help at the end of the project for a test on your own project.

Info for Phase A

Phase A

- Final recommendation ***(Conclusion card)***
  - Summarize results from assessment and testing
  - Generate specific and measurable design recommendations

  - This is the most important part of the activities of Phase A.
  - Be complete and exhaustive.
  - Connect each recommendation to some specific results from assessment and/or testing.
  - Make sure that recommendations are reasonable and implementable.

Phase B

3 Feasibility study

Perform a feasibility analysis for the development of your new idea.

3.1 Context of use

3.2 Scenarios

3.3 Personas

Phase B

3.1 Context of use

A document that describes accurately

- Who are the intended users and what are the tasks;
- What are the technical and environmental constraints.

This is for:

- Making sure that all factors affecting the use of the system have been identified before the start of design work;
- Providing a basis for designing usability tests

3.2 Scenarios

Little stories telling in detail how a certain user completes a personal goal realizing one or more task on the system.

- Goal and task relationship
- Decomposition of user tasks in actions (internal and external to the system)
- Identification of user's and system's operations
- Narrative of user actions, his goals and motivations to use the system
- DO NOT specify which features are used (the system is a black box)
- Establish timing estimates and success criteria for the scenario and for each task within it.
- Scenarios are useful for both common and critical situations

Phase B

3.3 Personas

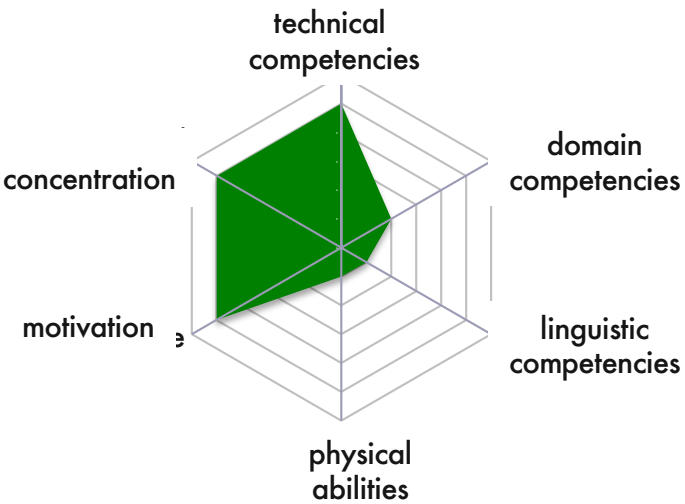
A dramaturgical and narrative approach to the project.

It tells a story, and assesses the interest and the credibility of the story itself, as you would evaluate a script for a movie

- Plots (scenarios)
- Characters (personas)
- Atmosphere (context)
- Plot (use cases)

Example 1: The new camper

Phase B

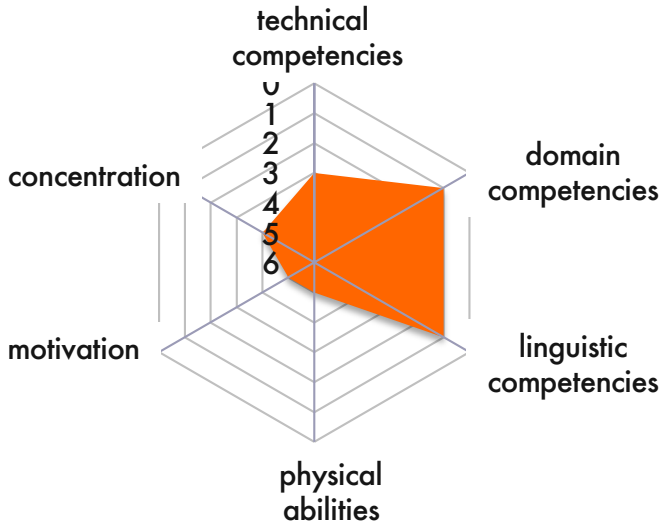


- Alberto is 41 years old, has wife, two children, a dog and a camper.
- Indeed he used to have a camper, which broke down after 10 years of service. He plans to get € 3,000 reselling it. He saw two chances, one is second hand, costing 13,000 € and a ne one costing 25,000 €. He wants to ask for a loan and return it in 5 years. Can he afford the new one?
- Alberto is a winemaker in a winery in the area. He is a direct person, educated, experienced and passionate, but hates computers and is very uncomfortable with them. He does not use them neither in his work nor for leisure and therefore does not own one. He uses the one in the administration office of the winery during the lunch break, but he is always in a hurry and has little patience.

He understands something about banks and loans.

Example 2: The new bedroom

Phase B



- Mahmur is 38 years old and is from Bangladesh. He arrived in Italy with his wife ten years ago, and soon after their daughters were born. He is a legal immigrant, works as a skilled worker for 1300 €/m.
- With his wife he decided to renovate the bedroom of their daughters. He wants to know how much he can receive by returning 200€/m in 3 years.
- Mahmur speaks low-level Italian. He speaks some English. He has no financial skills. He has been using computers and the Internet for years to talk with his family in Bangladesh, to read the newspapers of his country online, and participate in the Bengali immigration forums.
- He is kind, precise and meticulous, and loves things done well. He is never in a hurry, never loses his temper.

Competences and abilities

Phase B

- I propose here a simple model, divided into competences and abilities of exactly six basic user's characteristics (features)
- For each feature, you assign a numerical score (eg, from 1 to 5, with 1 a very low value and 5 a very high value),
- We then map the users' evaluation on a chart called strategy diagram.
- For example: Domain competence:
  - 5: Perfect mastery of terminology and procedures. He can find more different ways to deal with the same technical problem. High specialization
  - ...
  - 1: No knowledge of the purpose of the application. He must be lead step by step for each operation to perform on the application.

Competences and abilities

Phase B

- Technical competence:
  - nowadays no longer one-dimensional, given the complexity of today's technology from PCs to tablets, from Photoshop to Facebook, all different and without overlap.
  - Corner cases: **a young man who only uses a smartphone**, *a free-lance with a ten-years old PC, an elderly person with a young nephew helping him*.
- Domain competence:
  - the knowledge of practices, terminology, subtleties on the subject of the system.
  - Corner case: *a person exposed for the first time to the world of banks*.
- Language competence:
  - knowledge of the language of the system (e.g., Italian) and the ability to understand subtleties, innuendo, irony, sarcasm, different communication registers.
  - Corner case: *a foreigner in Italy for a few months*.

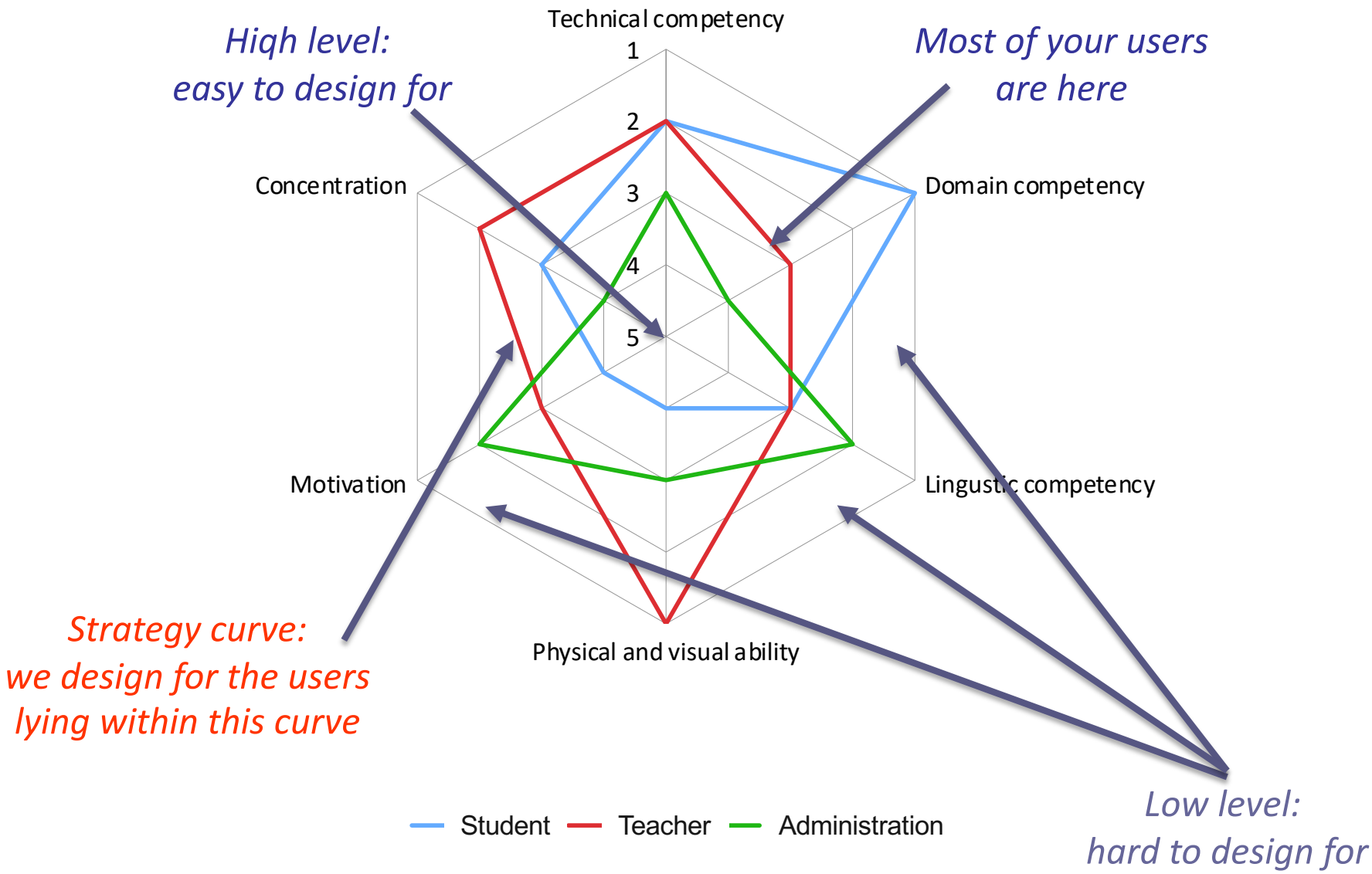
Competences and abilities

Phase B

- Physical Ability
  - the set of physical limitations, even partial and/or temporary, reducing the ability to use the device the system is running on.
  - Corner cases: *arthritis reducing mobility; presbyopia when not corrected by glasses*.
- Motivation
  - the set of motivations that cause the user to be interested in a task, to use a system for the task, to use THIS system for it.
  - Corner case: *a user who has already decided to use a competing system, and is just doing a quick check on yours*.
- Concentration (environmental distraction)
  - the ability or possibility, to provide adequate focus on the task.
  - Corner cases: *a person who accesses from an Internet point (therefore a lot of confusion), or standing in a crowded bus, or secretly from spouse or head office in his/her spare time.*

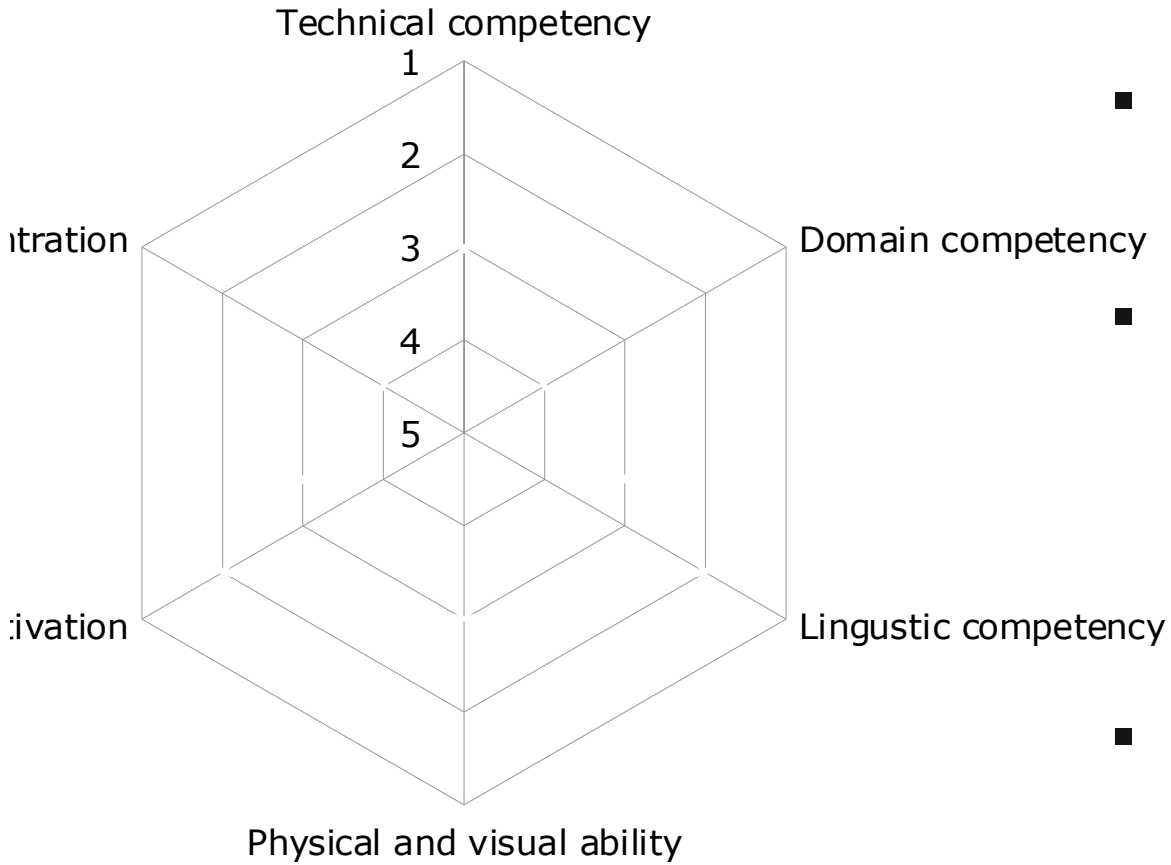
The C&A diagram

Phase B



The C&A diagram

Phase B

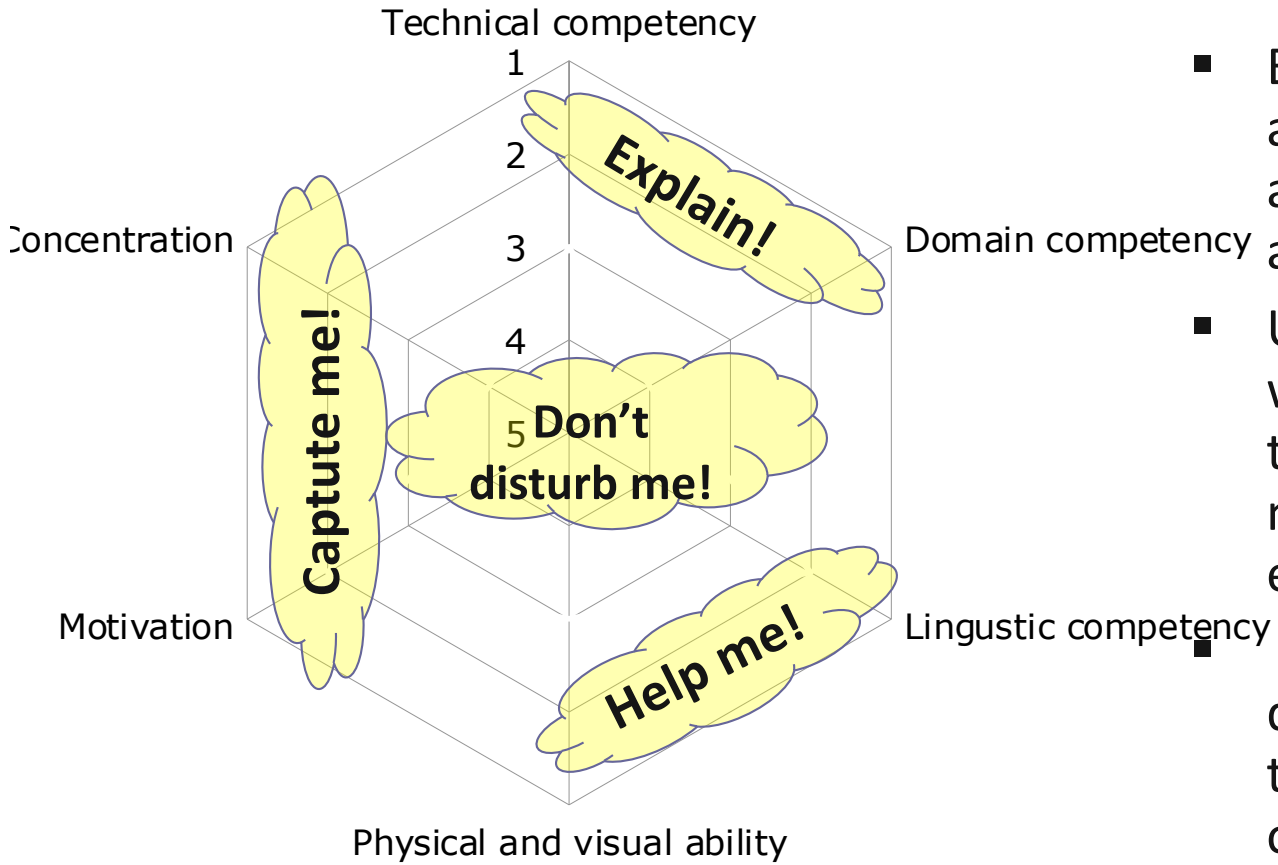


- Each character is associated with a C & A diagram. Each character has a score for each category.
- The total area of the chart represents the target user: each user occupies an area within the overall area.Outside the area there are no target users (that is: not relevant to the design)
- At the center there are people with high skills and abilities, near the edges people with lower ones.

An interesting persona is a borderline character (we must assume that designing for the average is easier than designing for the corner cases. Characters completely on the border (low competences and low abilities on all features) are extreme cases, intellectual toys. We strive to create personas that are very believable, and to give them credible characteristics.

Design impact of C&A diagram

Phase B



- Each area of the graph is associated with different types of actions to support the appropriate users.
- Users with high skills and abilities want efficient tools, complex, that help them achieve and maintain the state of flow in the execution of the task.
- users with low technical and domain competence requires textual and graphical explanation on the most obscure concepts.

Lack of linguistic or physical skils require additional aids in describing and/or activating the services.
Lack of motivations or concentration require additional ways to capture and maintaing the attention of the user, using even visceral-level aids or below the threshold of conscience.

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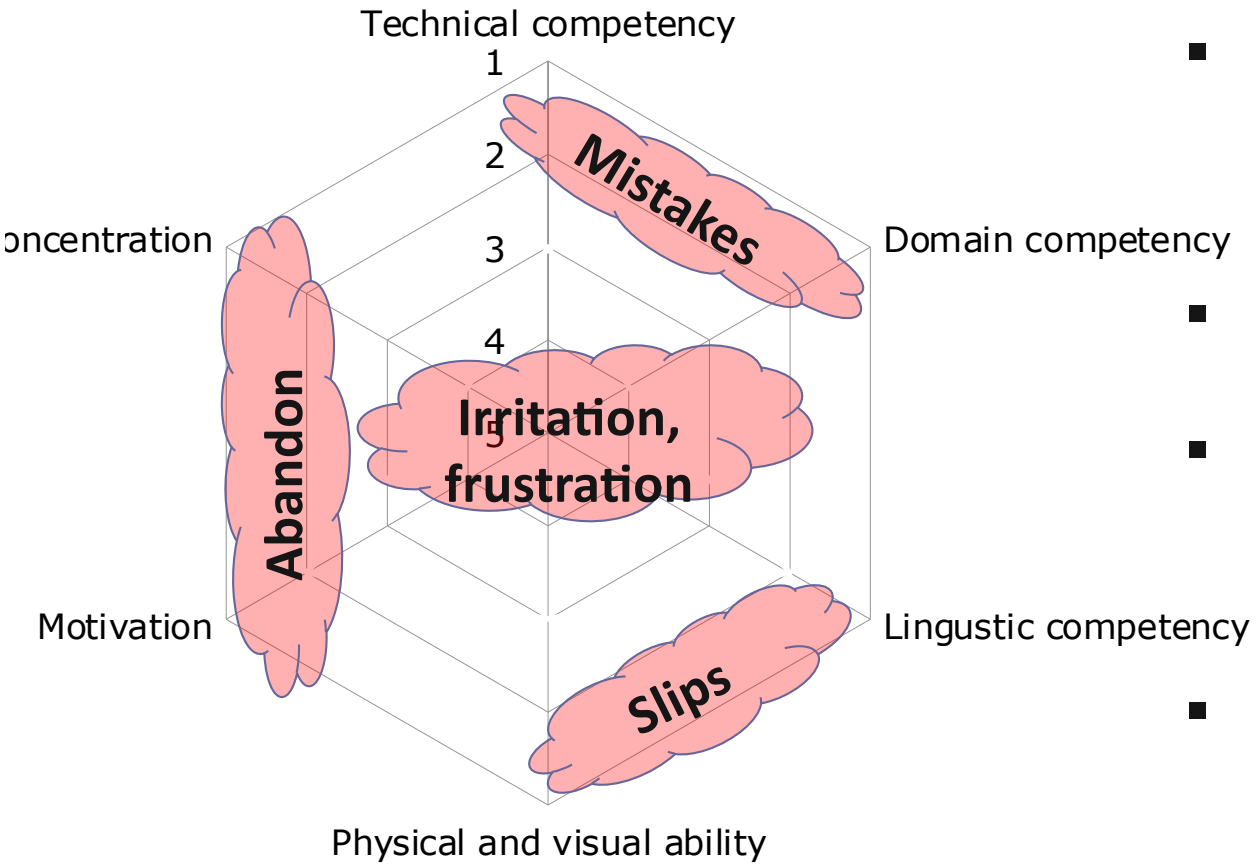
Design impact of C&A diagram

Phase B



- What is the risk if the value characterizing each dimension is not addressed correctly? It depends.
- Not solving competencies issues leads to making serious mistakes.
- Not solving language comprehension or physical fitness issues wil cause serious action errors, particularly slips.
- Not solving motivation or concentration issues will lead to users abandoning tasks halfway, which means losing a customer, a sale, etc.

Irritation and frustration are clear signs that a person knows what he/she wants, has good motivation, but the system is not helping. It is the result of not handling high competencies and skills.

Info for Phase B

Phase B

- ***Team card***
  - Repeat (identical) here
- ***Cast card***
  - General remarks of your user segments and the design choices you made to create the cast, including generated and discarded personas and scenarios.
- ***Persona card***
  - fill in all fields. Mainly text. Do not be too verbose.
  - Make sure your persona represents a specific segment and not others
  - Make sure you are creating believable characters full of details and peculiarities
  - These are the most important activities of Phase B. Be complete!
  - Connect each recommendation to some specific results from assessment and/or testing.
  - Make sure that personas are able to suggest non-obvious design ideas.

Phase C

4 Design proposal

Design your proposal

Before the design

4.0 Adoption of a design model

Design

4.1 Information Architecture

4.3 Interaction Design

4.4 Structure Blueprint

4.5 Wireframes

4.1 Information architecture

Phase C

Top-down design

- Create an overview of the system
- Every part of the system is progressively refined by adding more details to the design.
- It starts from the business goals and from there the users' goals and therefore the appropriate strategy to achieve these goal.

Bottom-up design

- Individual parts are described in detail
- The parts are connected to each other so as to form larger components, which are then interconnected to create a complete system.

Phase C

4.3 Interaction Design

- A design framework of the overall system
- It clarifies unambiguously:
  - what services are available
  - a clear hierarchical organization of the functions of the system.
- The content must be managed reasonably, they must be sorted, cataloged correctly so that information can be recalled effectively.

4.4 Blueprint

Phase C

Blueprints are drawings that define:

- The component for the organization of the content
- How these components are connected to each other.

It is usually difficult to represent a complex system with just one blueprint, so it is advisable to provide multiple perspectives for the information architecture.

It is also advisable to generate different versions depending on who will see your blueprint (programmers, clients, marketing department, etc.)

The blueprint must be complete, and include all the parts and connections that the implementers will have to inclue in the final product

Phase C

4.5 Wireframes

Drawings of the most fundamental parts of the page. A drawing for every screenful of the system.

You can (should) use a tool for wireframes

The wireframe must be complete, and include all the text, images, commands and labels that the user will need to see to complete the tasks

You do not have to create wireframes for the obvious passages (e.g., login), but DO include commands for accessing them in the wireframe you create.

Phase C

4.5 Wireframes (2)

- You can avoid useless details (e.g., the specific text shown when logging in) but DO NOT create overly generic interfaces.

- Since tone and messages are important for motivation, all parts of the interface that deliver tone and messages MUST BE FULLY DESIGNED AND DESCRIBED

- This includes: texts, images, layouts, colors and anything that can possibly be used to increase and strengthen motivation.

Info for Phase C

Phase C

- ***Information Architecture card***
  - General remarks of your design choices as per Information Architecture
- ***Blueprint*** *(Any form)*
  - Any tool can be used. be complete and thorough
- ***Wireframe*** *(Any form)*
  - Use your choice of tool.
  - But Lo-Fi sketches must be Lo-Fi, Hi-Fi drawings must be Hi-Fi
- ***Comparison*** *(Any form)*
  - An in depth before/after visualization of the same part of the application in the original tool and in your prototype
- ***Design card***
  - Some final reflections on how the design was carried out
  - Important: describe in detail how personas affected and influenced design

Phase D

5. Evaluation of the design

Evaluate your design with the same tools and methods you have used for the existing resources

5.1 Inspection

5.2 User testing

Phase D

5.1 Inspection of the project

The inspection phase takes place within the design team. For this reason, it is rather cheap technique (but also rather inaccurate).

There are three techniques relevant for this purpose:

- *Cognitive walkthrough*: a fictional and step by step execution of a task, and an empirical evaluation of the performance
- *Action analysis*: a quantitative analysis of specific actions that must be performed to perform a task.
- *Heuristic analysis* (or use of guidelines): the evaluation of a system based on a set of common-sense rules derived from past experiences.

Phase D

5.2 User testing

As in the user testing of the first phase, but applied to your system now.

Recommendation for testing (both here and before):

- Test with real users (no design fiction here: real people in flesh)
- Apply well known metrics in your tests. Do not invent measures
- Plan at least one *satisfaction-oriented test*: this is after all a user-experience design course, and satisfaction is the most important aspect of UUXD.
- Put the actual quantitative tables of tests in the documentation. It is important for me to be able to check your conceptual conclusions with the figures that can back up your opinions.
- Using different testers for current system and planned system means comparing absolute qualities of the two systems, while using the same testers means evaluating the improvement of the second system over the first one. Both are good choices but they are different choices. Be prepared to defend your choice. Make sure there is a consistent interval of time between the two tests, so that tester can "forget" details and impressions and opinions in the meantime.

Phase D

6 Conclusions and Recommendations

- What you wanted to get
- What has been achieved
- What are the data and the numbers that prove the result
- How to interpret this data
- What other activities could improve:
  - The reliability of the results obtained
  - Functional enrichment of the product
  - Further improvement of the usability of the product

Phase D

Info for Phase D

- ***Team card***
  - Repeat (identical) here
- ***Inspection card***
  - Inspection activities carried out by the team on the prototype
- ***Redesign user testing card***
  - Testing activities carried out by the team on the prototype
  - Very similar to preliminary user testing card
  - At least two users of the first session must be subjected to this session
  - *In case of ambiguous or unsatisfactory results of testing, plan another design and testing session!*
- ***Project conclusion card***
  - Some final reflections on how the whole activity was carried out, the errors you made and how you fixed them
  - Compare quantitatively and qualitatively your prototype with the original tool and stress you strong points and weaknesses.

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Final remarks

Please remember

- ***Find a tone and a message***
- Understand your product, understand your audience
- Be specific, be complex
- Make complete examples
- Use actual texts and actual images whenever they establish the tone and explain the message
- Handle background competencies! All these systems imply disconnect between required and available knowledge: make sure that help is not condescending and patronizing (users are treated as idiots), but rather helpful and constructive.

Frequent errors

- Some activities are to be carried out with syntethic users (personas), some others with real people in flesh and bones. Understand which is which and carry them out accordingly. Do not invent real users.
- USE PERSONAS APPROPRIATELY. Personas are not simply a check in the corresponding check box. They are tools to help focusing design and generate answers outside of your specific natural, spontaneous reactions. They are as good as you can separate their nature from your own.
- Be generic in your wireframes when the content is obvious or irrelevant: squares instead of images, squibbles instead of texts, lorem ipsum blocks instead of blocks are appropriate only when they really do not matter, otherwise it is ***way*** better to invent something plausible.
- Your tests matter. Fake tests are useless. Usually, summative tests are carried out by a different team than the designers, in order to minimize bias. You will do it yourself, usually with friends and family. Try to be as objective as possible. You will not be evaluated on how positive or negative are scores obtained in these tests, but in the overall likelyhood of these scores.

Deliverables

- I do not accept individual cards. Fill them in and compose a single document and generate a PDF.
- Attach a folder of additional material (tests, numerical tables, testing protocols, DISCARDED DESIGN CHOICES, etc.)
- Create a single ZIP file and upload it on virtuale.unibo.it in the appropriate area.

Before / after

- The documentation must include a comparison *Before / After* of your tool, so that it is clear which parts of the existing system have been affected and how.
- Be ready to defend your choices and how your decisions have improved on the existing system
- Existing system change frequently. It is very frustrating to find out that the system you based your "before" tests have changed when creating your documentation. Make screenshots while you see them.
- Your own system will evolve between early ideas to ready to test to after testing. Document all these phases in the Project Management documents

Conclusions

In this document we have discussed:

- Details of the completion of the end-of-course project for A.Y. 2024-25